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## Graduate education and enterprise innovation

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### ABSTRACT

Graduate education aims to cultivate high-level innovative talent and plays an important role in national innovation system. This study employs the difference-in-differences method to examine the impact of graduate education in Chinese universities on enterprise innovation. China implemented a graduate enrolment expansion policy in 2009, which is regarded as an exogenous policy shock in this study. This study finds that graduate enrolment expansion significantly promotes the innovation activities of high-tech enterprises. Such effect is more pronounced for non-state-owned enterprises and enterprises in developed regions. Furthermore, graduate enrolment expansion improves the education level of employees, and increases the number of research & development personnel of enterprises. This in turn upgrades human capital and promotes enterprise innovation. Our findings highlight the importance of a more labour-market-oriented higher education system, and provide implications for driving enterprise innovation through upgrading human capital.

### KEYWORDS

Graduate education;  
enterprise innovation;  
human capital; high-tech  
industrial

### JEL CLASSIFICATION

G30; I25; J24

## I. Introduction

After the global financial crisis, the Ministry of Education of China (MEC) largely expanded enrolment for graduate students, including Master's and Doctoral students, to ease the employment pressure and enhance human capital in 2009. The number of graduate students enrolled in China increased from 510,953 in 2009 to over 1.1 million in 2020. The theory of human capital believes that human capital accumulation is the source of economic growth, and the core of optimizing human capital is improving the intelligence level of workers through education (Schultz 1961). Graduate education is devoted to higher-order original research. Graduate students with professional knowledge and skills are the primary source of high-level research & development (R&D) personnel (Barnacle 2005). Moreover, the growth of the number of graduate students can effectively alleviate the shortage of high-level human capital of enterprises (Darley 2021).

Human capital is an important factor of driving enterprise innovation (Cao et al. 2022; Xu, Chen, and Fan 2023). Hiring R&D personnel with research experience from universities helps enterprises overcome technical bottlenecks (Herstad,

Sandven, and Ebersberger 2015). However, the effect of education expansion on human capital is greatly affected by national policies and economic situation (Buckner and Khoramshahi 2021; Wang, Wang, and Zong 2022). Based on the theory of human capital and Schumpeter's theory of innovation, this study examines the impact of China's graduate enrolment expansion on enterprise innovation from the perspective of human capital upgrading and provides policy implications for emerging economies to promote innovation through developing a more labour-market oriented higher education system.

## II. Methodology and data

This study selects China's non-financial A-share listed firms as the research sample. The graduate student data is collected from the *Educational Statistics Yearbook of China*. The patent and employee education data are obtained from the China Stock Market & Accounting Research Database and China Center for Economic Research Database, respectively. The MEC started to expand the graduate enrolment in 2009 in a large scale. Since the study period of graduate programmes is

normally 3 years in China and the growth rate of graduation peaked in 2012, 2012 is regarded as the year of policy shock. The MEC further expanded enrolment of graduate students in 2017 (mostly part-time graduate students) and in 2020 to mitigate the effect of COVID-19 on employment rate. To avoid the empirical results from being influenced by other policies, we choose 2008–2015 as the sample period. All continuous variables are winsorized at the 1% and 99% levels. The final sample consists of 11,716 firm-year observations.

The Difference-in-Differences (DID) model employed to investigate the impact of graduate education on enterprise innovation is shown as follows:

$$Innovation_{i,t+1} = \alpha_1 + \beta_1 Treat * Post_{i,t} + \delta_1 X_{i,t} + \theta_t + \mu_i + \varepsilon_{i,t} \quad (1)$$

$$Employee\_edu_{i,t} = \alpha_2 + \beta_2 Treat * Post_{i,t} + \delta_2 X_{i,t} + \theta_t + \mu_i + \varepsilon_{i,t} \quad (2)$$

$$RD\_personnel_{i,t} = \alpha_3 + \beta_3 Treat * Post_{i,t} + \delta_3 X_{i,t} + \theta_t + \mu_i + \varepsilon_{i,t} \quad (3)$$

$$RDP\_ratio_{i,t} = \alpha_4 + \beta_4 Treat * Post_{i,t} + \delta_4 X_{i,t} + \theta_t + \mu_i + \varepsilon_{i,t} \quad (4)$$

$$Employee\_edu = (1 \times Collegedegreeholders + 3 \times Bachelordegreeholders + 5 \times Postgraduatedegreeholders) \div Totalnumberofemployees \quad (5)$$

where  $\theta_t$  represents the time fixed effect,  $\mu_i$  represents the firm fixed effect, and  $\varepsilon_{i,t}$  is the error term.  $i$  and  $t$  represent firm and year, respectively. Referring to previous studies (Hou, Hu, and Yuan 2017; Wang 2021), *Innovation* represents the innovation outputs of a listed firm, including patent applications, invention patent applications, patent grants, and invention patent grants. The natural logarithms of one plus innovation outputs are used as the dependent variables. *Employee\_edu* is the employee's education level constructed using the weighting factor of 1:3:5 for vocational,

undergraduate and graduate education, as shown in Equation (5). *RD\_personnel* and *RDP\_ratio* are the natural logarithms of the total number of R&D personnel and the ratio of the total number of R&D personnel to the total number of employees, respectively. *Treat* takes the value of 1 if the company is from high-tech industry, and 0 otherwise. *Post* takes the value of 1 if the year is 2012 and later, and 0 otherwise.

### III. Empirical results

Table 1 reports the empirical results of the baseline regressions. Columns (1)–(4) are the effects of graduate enrolment expansion on the number of patent applications and grants. All coefficients of *Treat*×*Post* are significantly positive. The empirical results indicate that graduate enrolment expansion significantly promotes the innovation activities of high-tech enterprises. We further control the effect of R&D investment and re-estimate the DID models. As reported in Appendix, although the number of observations reduces from 11,716 to 7742, the regression results remain unchanged.

The validity of the DID method needs to meet the parallel trends assumption. As shown in Table 2, the coefficient of *Treat*×*Before* fails the significance test at the 10% level, indicating that before the implementation of the graduate enrolment expansion policy, there was no significant difference in the growth trend of the innovation outputs between high-tech and non-high-tech firms. This is in line with the parallel trend assumptions.

Table 3 reports the empirical results of the effects of graduate enrolment expansion on the education level of employees (*Employees\_edu*), the number of R&D personnel (*RD\_personnel*), and the ratio of R&D personnel (*RDP\_ratio*). All coefficients are significantly positive, which indicates that graduate enrolment expansion can effectively enhance the human capital of listed firms. Since human capital is one of the most important essentials of innovation (Protojerou, Caloghirou, and Vonortas 2017), graduate enrolment expansion policy can strengthen enterprises' capacity to innovate by upgrading human capital.

This study further investigates the impact of graduate education on the innovation of state-

**Table 1.** Patents and graduate education.

| Dep. Var.         | Apps<br>(1)          | Inv_apps<br>(2)      | Grants<br>(3)        | Inv_grants<br>(4)    |
|-------------------|----------------------|----------------------|----------------------|----------------------|
| Treat×Post        | 0.219**<br>(2.22)    | 0.141**<br>(2.17)    | 0.209*<br>(1.79)     | 0.214**<br>(2.26)    |
| Size              | 0.347***<br>(8.31)   | 0.290***<br>(12.44)  | 0.306***<br>(7.64)   | 0.197***<br>(6.54)   |
| Lev               | −0.145<br>(−1.07)    | −0.002<br>(−0.03)    | −0.148<br>(−1.15)    | −0.159*<br>(−1.76)   |
| Age               | 0.098**<br>(2.12)    | 0.134***<br>(4.04)   | 0.163***<br>(3.58)   | 0.110***<br>(2.86)   |
| ROA               | 0.629**<br>(2.51)    | 0.592***<br>(3.49)   | 0.096<br>(0.40)      | −0.447***<br>(−2.60) |
| BM                | −0.274***<br>(−3.07) | −0.246***<br>(−3.73) | −0.104<br>(−1.20)    | −0.120*<br>(−1.65)   |
| Cash              | −0.099<br>(−.90)     | −0.036<br>(−.40)     | −0.141<br>(−1.25)    | −0.054<br>(−0.56)    |
| Salespp           | −0.057*<br>(−1.92)   | −0.039**<br>(−2.22)  | −0.058**<br>(−2.12)  | −0.020<br>(−0.99)    |
| Turnover          | 0.172**<br>(2.14)    | 0.037<br>(.85)       | 0.151**<br>(1.97)    | −0.013<br>(−0.22)    |
| Exe_Share         | −0.471**<br>(−2.17)  | −0.100<br>(−.68)     | −0.344<br>(−1.64)    | 0.002<br>(0.01)      |
| Board             | −0.068<br>(−1.33)    | −0.014<br>(−.33)     | −0.045<br>(−0.88)    | −0.039<br>(−0.92)    |
| Top               | −0.607**<br>(−2.33)  | −0.569***<br>(−4.06) | −0.636**<br>(−2.55)  | −0.473**<br>(−2.51)  |
| Dual              | 0.046<br>(1.13)      | 0.037<br>(1.28)      | 0.015<br>(0.39)      | −0.020<br>(−0.62)    |
| Constant          | −4.434***<br>(−5.02) | −4.322***<br>(−9.13) | −4.097***<br>(−4.77) | −3.059***<br>(−4.92) |
| Firm fixed effect | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effect | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations      | 11716                | 11716                | 11716                | 11716                |
| R-squared         | 0.180                | 0.196                | 0.187                | 0.215                |

Variable definitions are reported in Appendix. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.

**Table 2.** Trend analysis.

| Dep. Var.                  | Apps<br>(1)       | Inv_apps<br>(2)   | Grants<br>(3)     | Inv_grants<br>(4) |
|----------------------------|-------------------|-------------------|-------------------|-------------------|
| Treat×Before <sup>−4</sup> | 0.118<br>(0.70)   | 0.017<br>(0.10)   | 0.100<br>(0.65)   | 0.079<br>(0.53)   |
| Treat×Before <sup>−3</sup> | −0.019<br>(−0.12) | 0.111<br>(0.70)   | −0.030<br>(−0.20) | 0.037<br>(0.30)   |
| Treat×Before <sup>−2</sup> | 0.017<br>(0.12)   | 0.105<br>(0.75)   | −0.056<br>(−0.43) | −0.021<br>(−0.20) |
| Treat×Before <sup>−1</sup> | 0.022<br>(0.17)   | 0.138<br>(1.11)   | −0.039<br>(−0.33) | 0.103<br>(1.08)   |
| Treat×Current              | 0.264*<br>(1.81)  | 0.302**<br>(2.14) | 0.138<br>(1.02)   | 0.153<br>(1.25)   |
| Treat×After <sup>1</sup>   | 0.279*<br>(1.81)  | 0.259*<br>(1.74)  | 0.303**<br>(2.09) | 0.263**<br>(2.10) |
| Treat×After <sup>2</sup>   | 0.306**<br>(1.97) | 0.328**<br>(2.22) | 0.105<br>(0.69)   | 0.241*<br>(1.95)  |
| Treat×After <sup>3</sup>   | 0.360**<br>(2.24) | 0.322**<br>(2.04) | 0.291*<br>(1.74)  | 0.184<br>(1.35)   |
| Control variables          | Yes               | Yes               | Yes               | Yes               |
| Firm fixed effect          | Yes               | Yes               | Yes               | Yes               |
| Year fixed effect          | Yes               | Yes               | Yes               | Yes               |
| Observations               | 11716             | 11716             | 11716             | 11716             |
| R-squared                  | 0.209             | 0.223             | 0.234             | 0.228             |

The t-statistics reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.

**Table 3.** The education level of employees, R&D personnel and graduate education.

| Dep. Var.         | Employees_edu    | RD_personnel    | RDP_ratio         |
|-------------------|------------------|-----------------|-------------------|
|                   | (1)              | (2)             | (3)               |
| Treat×Post        | .064**<br>(2.56) | .468*<br>(1.94) | .131***<br>(3.32) |
| Control variables | Yes              | Yes             | Yes               |
| Firm fixed effect | Yes              | Yes             | Yes               |
| Year fixed effect | Yes              | Yes             | Yes               |
| Observations      | 8099             | 1777            | 1665              |
| R-squared         | .160             | .397            | .291              |

The t-statistic reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.

owned enterprises (SOE) and non-state-owned enterprises (non-SOE). As shown in Table 4, the coefficients of *Treat×Before* are significantly positive in the non-SOE group, but insignificant in the SOE group. Before graduate enrolment expansion, it is difficult for non-SOEs to recruit graduate students, as SOEs usually provide more secured jobs and more stable remuneration. Therefore, the graduate enrolment expansion can better enhance the human capital and improve the innovation outputs of non-SOEs.

Moreover, this study examines whether the impact of graduate education on enterprise innovation varies across different regions. As reported in Table 5, the coefficients of *Treat×Post* are significantly positive in the group of more developed

region (i.e. eastern region). Graduate students in China usually prefer to pursue their career in more developed cities with resource advantages. Consequently, the effect of graduate enrolment expansion on innovation is more pronounced for high-tech firms located in eastern region. The central and western regions should emphasize the importance of a high-quality labour market in fostering domestic innovation.

#### IV. Conclusion

This study finds that the graduate enrolment expansion in China helps improve the innovation outputs of high-tech enterprises. The empirical evidence shows that graduate enrolment expansion improves the educational level of employees and the number of R&D personnel of enterprises. This further accelerates the accumulation of human capital and strengthens the innovation capability of enterprises. Furthermore, the impact of graduate education on enterprise innovation is more pronounced for non-SOEs and enterprises located in more developed regions in China. The policy implication of this study is that the state should cultivate a more labour-market oriented higher education system. Emerging countries can refer to China's practice and increase the level of human

**Table 4.** State-owned enterprises.

| Dep. Var.         | (1) State-owned enterprises |                   |                 |                 | (2) Non-state-owned enterprises |                  |                   |                   |
|-------------------|-----------------------------|-------------------|-----------------|-----------------|---------------------------------|------------------|-------------------|-------------------|
|                   | Apps                        | Inv_apps          | Grants          | Inv_grants      | Apps                            | Inv_apps         | Grants            | Inv_grants        |
| Treat×Post        | 0.061<br>(0.32)             | −0.022<br>(−0.11) | 0.080<br>(0.33) | 0.192<br>(0.91) | 0.285**<br>(2.45)               | 0.210*<br>(1.92) | 0.288**<br>(2.14) | 0.268**<br>(2.54) |
| Control variables | Yes                         | Yes               | Yes             | Yes             | Yes                             | Yes              | Yes               | Yes               |
| Firm fixed effect | Yes                         | Yes               | Yes             | Yes             | Yes                             | Yes              | Yes               | Yes               |
| Year fixed effect | Yes                         | Yes               | Yes             | Yes             | Yes                             | Yes              | Yes               | Yes               |
| Observations      | 4411                        | 4411              | 4411            | 4411            | 6942                            | 6942             | 6942              | 6942              |
| R-squared         | 0.210                       | 0.211             | 0.233           | 0.259           | 0.173                           | 0.190            | 0.171             | 0.197             |

The t-statistics reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.

**Table 5.** Region of enterprises.

| Dep. Var.         | (1) Eastern region |                   |                   |                   | (2) Central and western regions |                 |                   |                   |
|-------------------|--------------------|-------------------|-------------------|-------------------|---------------------------------|-----------------|-------------------|-------------------|
|                   | Apps               | Inv_apps          | Grants            | Inv_grants        | Apps                            | Inv_apps        | Grants            | Inv_grants        |
| Treat×Post        | 0.227**<br>(2.21)  | 0.159**<br>(2.32) | 0.270**<br>(2.10) | 0.261**<br>(2.52) | 0.228<br>(0.74)                 | 0.077<br>(0.22) | −0.076<br>(−0.29) | −0.029<br>(−0.13) |
| Control variables | Yes                | Yes               | Yes               | Yes               | Yes                             | Yes             | Yes               | Yes               |
| Firm fixed effect | Yes                | Yes               | Yes               | Yes               | Yes                             | Yes             | Yes               | Yes               |
| Year fixed effect | Yes                | Yes               | Yes               | Yes               | Yes                             | Yes             | Yes               | Yes               |
| Observations      | 8397               | 8397              | 8397              | 8397              | 3312                            | 3312            | 3312              | 3312              |
| R-squared         | 0.179              | 0.197             | 0.183             | 0.219             | 0.187                           | 0.201           | 0.198             | 0.213             |

The t-statistics reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.

capital by expanding the scale of graduate education, thereby strengthening the innovation capabilities of domestic enterprises.

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## Appendix

### Appendix A Variable definitions

| Variable      | Definition                                                                                                                                                                                                                                                                   |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apps          | The natural logarithm of one plus the number of patent applications in the year t.                                                                                                                                                                                           |
| Inv_apps      | The natural logarithm of one plus the number of patent applications for invention in the year t.                                                                                                                                                                             |
| Grants        | The natural logarithm of one plus the number of patent granted in the year t.                                                                                                                                                                                                |
| Inv_grants    | The natural logarithm of one plus the number of patent granted for invention in the year t.                                                                                                                                                                                  |
| Treat         | Dummy variable equals one if a firm belongs to the information transmission, software and information technology services according to the industry classification standard of listed companies issued by China Securities Regulatory Commission in 2012 and zero otherwise. |
| Post          | Dummy variable equals one if year is after 2012 and zero otherwise.                                                                                                                                                                                                          |
| Size          | The natural logarithm of total assets in the year t.                                                                                                                                                                                                                         |
| Lev           | The total liabilities divided by total assets in the year t.                                                                                                                                                                                                                 |
| Age           | The natural logarithm of firm age.                                                                                                                                                                                                                                           |
| ROA           | The return divided by total assets in the year t.                                                                                                                                                                                                                            |
| BM            | The book value of total assets divided by the market value in the year t.                                                                                                                                                                                                    |
| Cash          | The cash holding divided by total assets in the year t.                                                                                                                                                                                                                      |
| Salespp       | The natural logarithm of the operating income divided by the total number of employees in the year t.                                                                                                                                                                        |
| Turnover      | The operating income divided by total assets in the year t.                                                                                                                                                                                                                  |
| Exe_share     | The percentage of share holding by the executive in the year t.                                                                                                                                                                                                              |
| Board         | The natural logarithm of the number of directors in the year t.                                                                                                                                                                                                              |
| Top           | The percentage of share holding by the largest shareholder in the year t.                                                                                                                                                                                                    |
| Dual          | The dummy variable equals one if a firm's CEO also holds the position of board chair in the year t and zero otherwise.                                                                                                                                                       |
| Employees_edu | The variable constructed by the ratio of 1:3:5 for employees with different academic degrees in the year t.                                                                                                                                                                  |
| RD_personnel  | The natural logarithm of the company's R&D personnel in the year t.                                                                                                                                                                                                          |
| RDP_ratio     | The percentage of R&D personnel to total employees in the year t.                                                                                                                                                                                                            |

### Appendix B The number of graduate students and enterprise patents

| Dep. Var.         | Apps<br>(1)        | Inv_apps<br>(2)    | Grants<br>(3)      | Inv_grants<br>(4)  |
|-------------------|--------------------|--------------------|--------------------|--------------------|
| Graduate_stu      | 0.252***<br>(4.10) | 0.230***<br>(4.14) | 0.243***<br>(3.94) | 0.228***<br>(4.06) |
| Firm fixed effect | Yes                | Yes                | Yes                | Yes                |
| Year fixed effect | Yes                | Yes                | Yes                | Yes                |
| Observations      | 22569              | 22569              | 22569              | 22569              |
| R-squared         | 0.379              | 0.238              | 0.384              | 0.243              |
| Master_stu        | 0.218***<br>(3.41) | 0.201***<br>(3.49) | 0.207***<br>(3.23) | 0.197***<br>(3.39) |
| Firm fixed effect | Yes                | Yes                | Yes                | Yes                |
| Year fixed effect | Yes                | Yes                | Yes                | Yes                |
| Observations      | 22569              | 22569              | 22569              | 22569              |
| R-squared         | 0.379              | 0.238              | 0.384              | 0.243              |
| Doctor_stu        | 0.165***<br>(2.73) | 0.166***<br>(3.02) | 0.167***<br>(2.75) | 0.171***<br>(3.17) |
| Firm fixed effect | Yes                | Yes                | Yes                | Yes                |
| Year fixed effect | Yes                | Yes                | Yes                | Yes                |
| Observations      | 22569              | 22569              | 22569              | 22569              |
| R-squared         | 0.379              | 0.239              | 0.384              | 0.243              |

This table provides the estimation results from OLS regressions testing the effect of number of Master and PhD on patents from 2007 to 2021. Explanatory variables are Graduate\_stu, Master\_stu and Doctor\_stu, representing the natural logarithm of one plus the number of total graduate students, the number of master's students, and the number of doctoral students, respectively. In Column (1)-(4), the dependent variables are Apps, Inv\_apps, Grants and Inv\_grants, respectively. Variable definitions are reported in Appendix A. The regressions control for year and firm fixed effects. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.

## Appendix C Robustness test

| Dep. Var.         | Apps<br>(1)          | Inv_apps<br>(2)      | Grants<br>(3)         | Inv_grants<br>(4)    |
|-------------------|----------------------|----------------------|-----------------------|----------------------|
| Treat×Post        | 0.129*<br>(1.80)     | 0.004<br>(0.05)      | 0.163**<br>(2.39)     | 0.146**<br>(2.33)    |
| R&D_inv           | 0.074***<br>(4.88)   | .088***<br>(4.46)    | 0.104***<br>(7.28)    | 0.076***<br>(5.77)   |
| Constant          | −7.890***<br>(−1.39) | −7.850***<br>(−6.62) | −7.955***<br>(−11.00) | −5.443***<br>(−8.25) |
| Firm fixed effect | Yes                  | Yes                  | Yes                   | Yes                  |
| Year fixed effect | Yes                  | Yes                  | Yes                   | Yes                  |
| Observations      | 7742                 | 7742                 | 7742                  | 7742                 |
| R-squared         | 0.203                | 0.220                | 0.208                 | 0.238                |

R&D\_inv is the newly added control variable, the natural logarithm of the number of R&D investment in the year  $t$ . Variable definitions are reported in Appendix A. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. \*\*\*, \*\*, and \* denote significance at the 1%, 5% and 10% level, respectively.